# BirdNET → ONNX

*From an untrusted benchmark to a model we can actually prove works*

Coleman · ONNX Export, Phases 0–5

# Why We Did This

We had old benchmarks that were untrusted. This was a chance to reproduce BirdNET in a portable, ONNX format — and actually PROVE it behaves identically to the original, not just hope so.

### The old numbers were suspect

Wrong measurement tools made FP16 look worse than it was.

### Not retraining — re-proving

Same math, new format. The whole job is: does it still sound the same?

### The Goal

## Bit-for-bit

the same predictions, in a portable format ONNX Runtime that can run anywhere

# The Wall, and the "Aha"

### The conversion tool hit a wall

BirdNET computes its spectrogram using a math operation the standard TensorFlow→ONNX converter simply didn’t know how to translate. It failed the same way at every setting we tried.

### The insight that unlocked it

That "unsupported" math operation is secretly just a fixed matrix multiplication in disguise. So instead of fighting the tool, we swapped in the equivalent multiplication — exact same math, ops the converter already understood.

*Verified numerically before converting anything — no guessing.*

# The Detective Story

*When expanded to all six recorders, agreement dropped hard on some of them. The instinct is to blame the model. We didn't.*

**Checked inputs**

Bit-for-bit identical to the reference pipeline. Not that.

**Checked the model**

Original TensorFlow model agreed with the ONNX model exactly. Not the conversion.

**Found the real cause**

The audio clips themselves were cut to a slightly different length than the reference expected — a data mismatch, not a model bug.

**37,596** clips validated, once the data issue was untangled

**100.000%** agreement with the original model

# Where We Are Now

### FP16 is validated — and it's nearly free

Half the file size, 99.85% agreement with full precision. We even caught a subtle measurement trap along the way — a result THAT good was questionable, and confirmed exactly why the number came out that way before trusting it.

**99.854%**

top-1 agreement, half the size

*every disagreement was two owls being confused for each other — nothing broken*

### What's next

- Confirm target hardware with Kyle — it decides how we read this number
- Measure speed & memory (Phase 6)
- Fix the pruning plan's label mismatch before building on it
- Package everything so it's reproducible by anyone on the team